

Interference Characterization and Localization Framework for FMCW mmWave Radars

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Abstract—Mutual interference (MI) among automotive mmWave FMCW radars degrades detection reliability and Doppler coherence, which poses a challenge for autonomous systems. This study analyzes MI using two Texas Instruments IWR6843AOP radars operated under controlled slope-mismatch conditions. A MATLAB-based analyzer was developed to compare baseline and interference data through range-SNR histograms and clustering. After adaptive threshold calibration, distinct interference clusters were successfully identified, demonstrating improved detection sensitivity. The analyzer's results were further validated through a velocity-domain evaluation using the Average Directional Variance (ADV) metric, which quantified velocity decoherence caused by interference. The findings establish a foundation for future interference-based localization research in automotive radar systems.

Index Terms—Automotive radar, mutual interference, mmWave, FMCW, velocity coherence, Average Directional Variance (ADV), adaptive thresholding, interference localization.

I. INTRODUCTION

Frequency-Modulated Continuous-Wave (FMCW) radars, particularly those operating in the millimeter-wave (mmWave) spectrum (60–64 GHz and 77–81 GHz), have become central to modern sensing technologies due to their high range accuracy, compact design, and robustness under adverse weather conditions [1], [2]. These advantages have driven their widespread adoption in applications such as industrial automation, gesture recognition, and most notably, Advanced Driver Assistance Systems (ADAS) and autonomous driving [3], [4]. In these systems, multiple radars often operate simultaneously around a single vehicle, forming dense radar environments.

However, as radar deployment density increases, the problem of Mutual Interference (MI) has emerged as a critical limitation in FMCW systems [5], [6]. When multiple radars operate concurrently within overlapping time–frequency windows, the victim radar receives unwanted energy from other transmitters. This interference elevates the noise floor, reduces detection sensitivity and more critically introduces non-coherent distortions that appear as ghost targets or corrupted Doppler signatures [2]. The resulting degradation in velocity coherence directly undermines object tracking reliability in ADAS and autonomous perception systems.

While considerable research has focused on understanding and mitigating radar interference [7]–[9], most existing studies emphasize signal domain mitigation developing algorithms for waveform scheduling, adaptive filtering, or interference suppression at the receiver [10], [11]. In contrast, fewer studies have experimentally investigated how interference manifests in the processed radar domain, particularly in range–velocity–SNR data where real-time perception and tracking operate. Understanding these manifestations is essential for developing reliable diagnostic and detection tools.

This project addresses that gap by experimentally characterizing the effects of mutual interference using two Texas Instruments IWR6843AOP EVM radars operating in the 60–64 GHz band. Instead of applying mitigation, the study focuses on diagnosing interference through the radar's velocity-domain signatures. A MATLAB-based analysis framework was developed to quantify interference-induced decoherence using a novel metric Average Directional Variance (ADV) which measures the loss of Doppler consistency across frames.

Through comparative analysis of baseline (interference-free) and interference-present data, the study demonstrates how slope mismatch between chirp sequences produces measurable degradation in signal quality, cluster integrity, and velocity stability. These findings form a foundation for future research aimed at localizing the interference source using angular and temporal correlation methods, thereby bridging the gap between interference diagnosis and spatial awareness in automotive radar systems.

II. LITERATURE STUDY

A. Overview of FMCW Radar Principles

Frequency-Modulated Continuous Wave (FMCW) radar systems are widely adopted for short-range sensing applications such as automotive, industrial, and motion detection [12], [13]. Unlike pulsed radars, FMCW radars continuously transmit a frequency-modulated waveform known as a *chirp*.

A linear FMCW chirp can be expressed as:

$$f(t) = f_c + \mu t, \quad 0 \leq t \leq T_c \quad (1)$$

where f_c is the carrier frequency, T_c is the chirp duration, $\mu = \frac{B}{T_c}$ is the chirp slope, and B is the total bandwidth [12].

When the transmitted signal reflects off a target and returns after a delay $\tau = \frac{2R}{c}$, the received signal is mixed with the transmitted signal to produce an intermediate frequency (IF), also known as the beat frequency:

$$f_b = \mu\tau = \frac{2\mu R}{c}. \quad (2)$$

This beat frequency encodes the range R of the target. Performing a Fast Fourier Transform (FFT) on the IF signal enables extraction of distance and velocity information.

B. Radar Interference Mechanisms

In dense radar environments, such as multiple vehicles operating within the same frequency band, simultaneous FMCW transmissions can overlap in both time and frequency. This phenomenon, known as *mutual interference* (MI), produces *ghost targets*, *elevated noise floors*, and *false detections* [2], [11].

Assume two radars transmit:

$$s_A(t) = \cos \left[2\pi \left(f_{cA}t + \frac{1}{2}\mu_A t^2 \right) \right] \quad (3)$$

$$s_B(t) = \cos \left[2\pi \left(f_{cB}t + \frac{1}{2}\mu_B t^2 \right) \right] \quad (4)$$

After mixing, the interference term becomes:

$$f_{\text{int}}(t) = (f_{cA} - f_{cB}) + (\mu_A - \mu_B)t. \quad (5)$$

The difference in slope ($\mu_A - \mu_B$) determines whether the interference produces narrowband ghost targets or broadband noise:

- If $\mu_A \approx \mu_B$, a nearly constant beat frequency appears—potentially interpreted as a false target.
- If $\mu_A \neq \mu_B$, the beat frequency sweeps over time, creating broadband noise and raising the noise floor.

C. Effect of Chirp Slope Variation

1) *Chirp Slope and Range Relationship*: The chirp slope directly affects range estimation through:

$$R = \frac{cf_b}{2\mu}. \quad (6)$$

Hence, a variation in μ influences how beat frequency maps to distance [5].

2) *Interference Due to Slope Mismatch*: When two FMCW radars operate in close proximity and transmit frequency ramps with slightly different slopes, a phenomenon known as slope-mismatch interference occurs [2], [5]. Let the chirp slopes be μ_A (victim) and μ_B (interferer), with difference:

$$\Delta\mu = \mu_B - \mu_A. \quad (7)$$

The instantaneous interference frequency is:

$$f_{\text{int}}(t) = |(f_{cB} - f_{cA}) + (\mu_B - \mu_A)t|. \quad (8)$$

When $\Delta\mu \neq 0$, periodic intersections of the frequency ramps generate transient interference spikes within the radar's IF or ADC bandwidth. If $f_{\text{int}}(t)$ falls inside the ADC band, this additional energy appears indistinguishable from genuine target reflections, producing bursts of noise or spurious peaks in the Range–Doppler map. This interference is most pronounced when

$$|\Delta\mu| \leq \frac{B_{\text{ADC}}}{T_{\text{chirp}}}, \quad (9)$$

where B_{ADC} is the ADC bandwidth and T_{chirp} is the chirp duration.

D. Velocity-Domain Analysis and Coherence

In FMCW radar, Doppler processing enables estimation of a target's radial velocity by measuring the phase shift across successive chirps. The Doppler frequency f_D is related to the target velocity v as:

$$f_D = \frac{2v}{\lambda}, \quad (10)$$

where λ is the wavelength of the transmitted signal [12]. A second-stage FFT along the slow-time dimension converts phase variations into velocity estimates, forming a two-dimensional Range–Doppler map.

Under interference-free conditions, true targets exhibit coherent and stable velocity patterns across frames. Mutual interference, however, introduces non-coherent signal components and phase distortions, leading to velocity fluctuations, smeared Doppler peaks, and false moving detections [2], [9]. Recent research emphasizes the importance of quantifying velocity coherence as a diagnostic indicator of interference-induced corruption. Metrics such as velocity variance or frame-to-frame Doppler stability have been proposed to measure coherence loss [7], [14], [15]. Building upon this foundation, the present study employs the *Average Directional Variance* (ADV) metric to evaluate velocity decoherence, offering a practical means of detecting and characterizing mutual interference within processed radar data.

E. Clutter Removal Techniques

In FMCW radar signal processing, static clutter removal is a fundamental step to distinguish stationary background reflections (e.g., walls, floors, and large metallic structures) from dynamic targets such as humans or vehicles. Clutter primarily manifests as strong DC and low-Doppler components in the Range–Doppler map, masking moving targets and reducing detection sensitivity [1], [12]. The objective of clutter suppression is therefore to attenuate static reflections while preserving motion information.

1) *Static Background Subtraction*: One of the simplest and most effective approaches for clutter removal is the subtraction of an averaged background profile. Over several radar frames, the mean magnitude of each range bin is computed and subtracted from the current frame:

$$S_{\text{clean}}(r, t) = S(r, t) - \frac{1}{N} \sum_{i=1}^N S(r, t_i), \quad (11)$$

where $S(r, t)$ represents the complex signal at range bin r and time frame t , and N is the number of frames used for background estimation [16]. This technique efficiently removes strong static components but may also suppress slowly moving targets with very low Doppler signatures.

2) *Moving Target Indication (MTI) Filtering*: Moving Target Indication (MTI) filtering operates in the slow-time (Doppler) domain and removes the zero-Doppler energy component. The simplest form of MTI is a first-order high-pass filter defined as:

$$y[n] = x[n] - x[n - 1], \quad (12)$$

where $x[n]$ and $y[n]$ denote the input and output signal sequences, respectively. While effective for basic clutter suppression, more advanced implementations employ multi-tap FIR or IIR filters to achieve steeper Doppler selectivity and greater rejection of stationary clutter [17]. MTI filters are commonly applied prior to Doppler FFT processing to enhance the contrast between static and dynamic returns.

3) *CFAR-Based Clutter Suppression*: Constant False Alarm Rate (CFAR) detection dynamically adjusts the detection threshold according to local noise and clutter statistics, maintaining a uniform false alarm probability across varying environments [18]. The most common implementation, Cell-Averaging CFAR (CA-CFAR), defines the threshold as:

$$T = \alpha \times \frac{1}{2N} \sum_{i=1}^N (P_{i,\text{left}} + P_{i,\text{right}}), \quad (13)$$

where α is a scaling factor determined by the desired false alarm rate, and $P_{i,\text{left}}$ and $P_{i,\text{right}}$ are the reference cells used to estimate the background power [1], [16].

The reviewed studies consistently indicate that slope-mismatch interference and Doppler decoherence are major sources of performance degradation in FMCW radar systems. Mutual interference elevates the noise floor, introduces false targets, and disrupts velocity coherence, while clutter suppression methods can enhance visibility but must be applied cautiously to avoid altering interference characteristics. These theoretical insights form the basis of the experimental configuration described in the following section, where two IWR6843AOP radars were used to reproduce and analyze mutual interference behavior and evaluate velocity-domain coherence metrics.

III. METHODOLOGY

A. Experimental Setup

The experiment was conducted in an indoor environment with controlled ambient conditions. Two IWR6843AOP EVMs were positioned at fixed, non-collinear locations to ensure overlapping fields of view. One radar was configured as the *observer*, operating in standard single-chip FMCW mode to capture reflections from surrounding objects and the moving subject. The second radar, denoted as the *interferer*, was programmed with independently generated chirp sequences designed to overlap partially in time and frequency with the

observer radar's chirps, thereby inducing controlled mutual interference [11].

A single human subject (the experimenter) was used as a dynamic target, walking slowly through the shared field of view to produce measurable Doppler shifts. This configuration ensured realistic motion signatures while allowing consistent and repeatable interference exposure conditions.

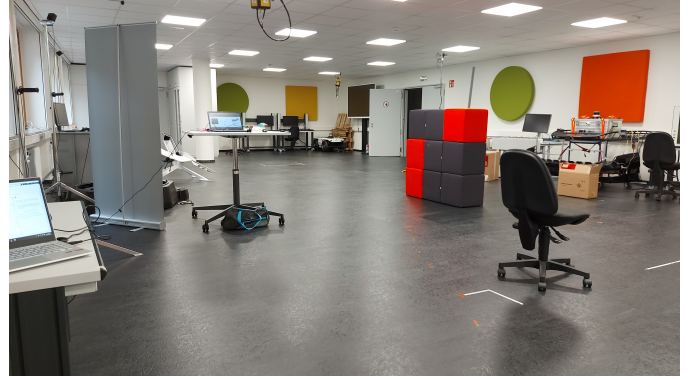


Fig. 1. Experimental setup showing the observer radar, interferer radar and static objects.

B. Data Acquisition and Parsing

Data acquisition was performed using two Texas Instruments IWR6843AOP EVMs connected via UART serial interface. Both radars were configured using identical chirp parameters, ensuring synchronized transmission characteristics while allowing controlled slope variations between test sets. For each experiment, two TLVs (Type-Length-Value packets) were recorded:

- **TLV 1**: Position and velocity data, providing 3D Cartesian coordinates (x, y, z) and corresponding radial velocity v for each detected point.
- **TLV 7**: Signal-to-noise ratio (SNR) and noise-level data, representing signal strength and background noise conditions for each detection.

The raw TLV data were parsed and stored using a customized MATLAB parser adapted from TI's mmWave SDK [11]. These structured arrays formed the foundation for subsequent ADV and clustering analyses.

C. Data Analysis and Metric Evaluation

Post-processing was carried out using a MATLAB-based analyzer designed to compare baseline (non-interfered) and interference datasets. The objective was to quantify degradation in velocity coherence and signal quality through:

- Computation of False Positive Rate (FPR) and False Discovery Rate (FDR) [7].
- DBSCAN-based clustering to identify object groupings in spatial data [19].
- Range-SNR histograms and statistical metrics to evaluate distribution broadening.

D. ADV-Based Velocity Profile Analysis

An additional MATLAB tool was developed to visualize the velocity profile per frame for both baseline and interference conditions. This analysis plotted instantaneous velocity distributions across frames, revealing the loss of Doppler consistency due to interference. The observed random fluctuations validated the *Average Directional Variance (ADV)* metric as a quantitative measure of velocity decoherence [14], [15].

E. Summary of Methodology

In summary, the methodology integrates:

- 1) Dual-radar experimental setup using IWR6843AOP EVMs [11].
- 2) Serial-based TLV data acquisition and MATLAB-based analysis.
- 3) Statistical evaluation and DBSCAN clustering [19].
- 4) ADV-based velocity profiling for coherence quantification [14], [15].

Together, these steps form a diagnostic pipeline for identifying and characterizing mutual interference artifacts in FMCW radar data, demonstrating the effectiveness of velocity-domain analysis [1], [2].

IV. RESULTS AND DISCUSSION

This section presents the analysis of mutual interference effects observed under the main radar configuration setup. The experiment was conducted using two IWR6843AOP EVM radars—one configured as the *observer* and the other as the *interferer*. Both devices were configured to operate at 60 GHz with the parameters summarized below:

- Baseline chirp slope: 3 MHz/ μ s
- Interferer chirp slope: 7 MHz/ μ s
- Total frames: 500
- Static clutter removal: Disabled
- Range resolution: 0.044 m
- Maximum radial velocity: 1 m/s
- Doppler resolution: 0.13 m/s
- Frame duration: 100 ms

This configuration was selected to provide high range precision while ensuring sufficient Doppler sensitivity for slow-moving targets. The interferer radar transmitted chirps with a steeper slope to create overlapping frequency ramps and induce controlled mutual interference within the observer radar's reception window. Each experiment was recorded for 500 frames using a MATLAB-based serial data parser. The primary goal of this analysis was to identify and quantify interference effects through object clustering, performance metrics, and velocity profiling.

Performance Analysis

The first step involved evaluating baseline and interference conditions using the initial MATLAB analyzer. Table I summarizes the key performance metrics, including total detections, clustering statistics, and calculated error rates.

The initial results indicate that while the total number of detections increased significantly under interference, no

TABLE I
PERFORMANCE METRICS USING INITIAL ANALYZER (NO THRESHOLDING APPLIED).

Parameter	Baseline Data	Interference Data
Total Raw Detections	1171	1944
Estimated Objects (Clustered)	9	9
Static Objects	4	4
Moving Objects	5	5
Interference Objects	0	0
Noise Detections	158	148
False Positive Rate (FPR)	13.49%	7.61%
False Discovery Rate (FDR)	0.00%	0.00%

separate interference clusters were detected. The unchanged number of estimated objects suggests that the analyzer's static thresholding approach was insufficient to capture the transient nature of non-coherent interference artifacts.

Histogram and SNR Observations

To further examine interference behavior, range-SNR histograms were plotted for both baseline and interference conditions. The interference histogram exhibited a noticeable broadening of the SNR distribution and a shift toward lower peak magnitudes, reflecting degradation in detection confidence and the presence of overlapping chirps. These observations confirm the impact of external radar signals on signal quality and clutter density.

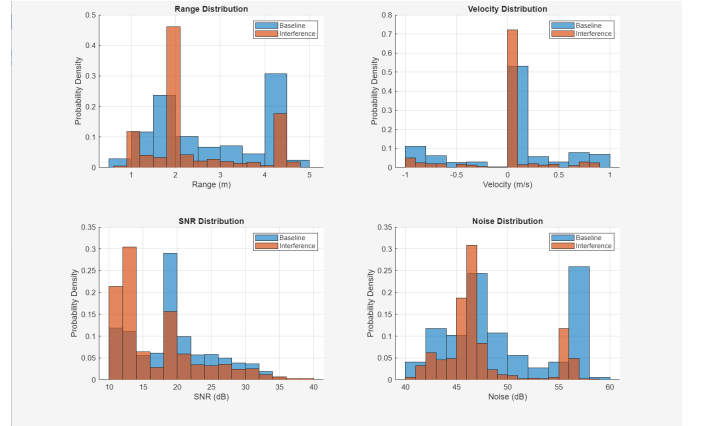


Fig. 2. Comparison of range, SNR, Velocity and Noise histograms for baseline and interference conditions.

Clustering Analysis

Figure 3 shows the clustering outcome obtained from the baseline and interference datasets using the initial analyzer. Although the total number of detections increased under interference, all points were classified within existing object clusters, resulting in no new interference-specific groupings. This behavior reflects the limitation of fixed detection thresholds and static clustering parameters in separating valid detections from spurious ones.

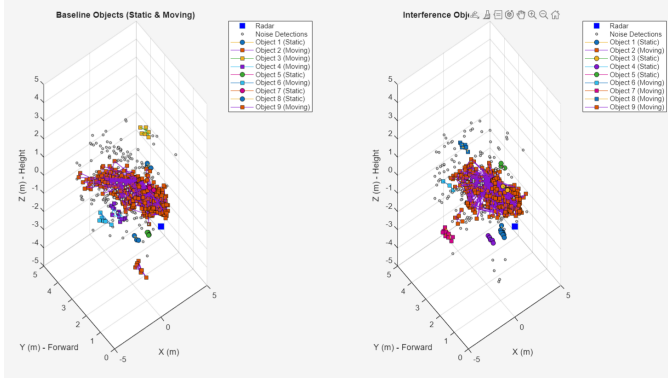


Fig. 3. Initial clustering plots showing baseline and interference datasets without threshold optimization.

Threshold Optimization and Enhanced Analyzer Results

To enhance separation, a MATLAB-based threshold estimation was performed using data where both radars had identical chirp slopes but different total frame counts. This produced a calibrated range threshold, as illustrated in Figure 4, used to refine the clustering parameters.

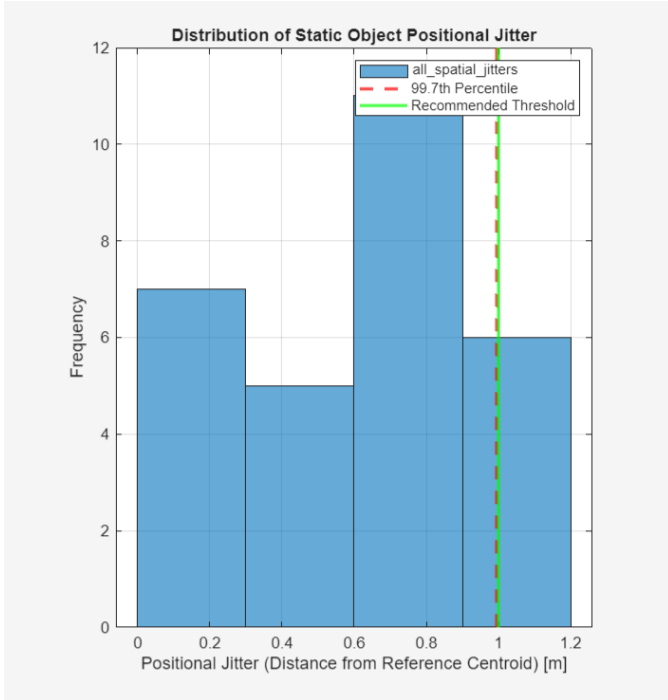


Fig. 4. Threshold estimation plot derived from slope-matched data, identifying optimal range threshold for interference detection.

After applying the threshold, updated clustering plots (Figure 5) revealed distinct interference clusters that were previously unidentifiable, confirming that threshold optimization significantly improved interference detection sensitivity. The refined version utilized kinematic-spatial matching and dynamic detection limits to better identify anomalous clusters

caused by interference. This approach was tested on the same dataset (500 frames) using the same radar configuration.

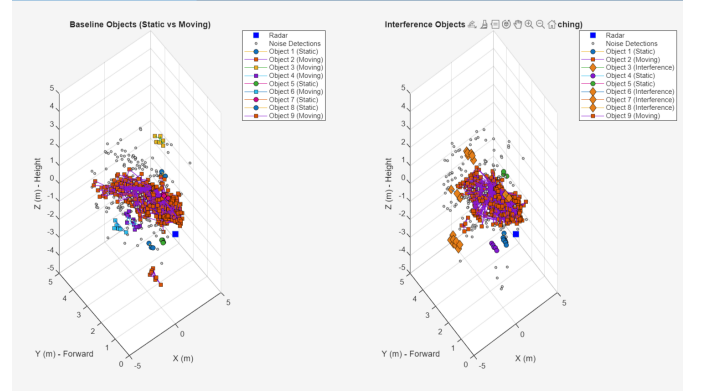


Fig. 5. Updated clustering plots showing newly identified interference clusters after threshold optimization.

Note: Matching parameters used for interference data — Feature Weights [R, V, A]: [1.0, 5.0, 0.1], Match Threshold (Weighted Dist): 1.00.

Threshold Optimization and Enhanced Interference Detection

The initial analysis (Table I) showed comparable detection performance between the baseline and interference datasets. Both cases produced nine clustered objects, with no interference objects identified. The False Discovery Rate (FDR) remained at 0%, suggesting that the system failed to distinguish interference-induced anomalies from background detections. While total raw detections were higher in the interference condition, the clustering algorithm was unable to separate them as distinct interference components. After applying the threshold, the updated clustering plots (Figure 5) revealed distinct interference clusters that were previously unidentifiable, confirming that threshold optimization significantly improved interference detection sensitivity. The updated analyzer also produced more representative metrics, as shown in Table II, reflecting improved separation between valid object detections and interference-induced anomalies.

TABLE II
PERFORMANCE METRICS USING UPDATED ANALYZER WITH THRESHOLD OPTIMIZATION.

Parameter	Baseline Data	Interference Data
Total Raw Detections	1171	1944
Estimated Objects (Clustered)	9	9
Static Objects	4	3
Moving Objects	5	2
Interference Objects	0	4
Noise Detections	158	148
False Positive Rate (FPR)	13.49%	7.61%
False Discovery Rate (FDR)	0.00%	44.44%

Compared to the initial analyzer, the updated framework successfully differentiates interference-induced detections from normal object responses. The introduction of adaptive thresholding enabled the identification of four interference

clusters in the interference dataset, which were absent in the baseline analysis. This improvement demonstrates the system's ability to isolate spurious returns caused by overlapping chirps.

The increased False Discovery Rate (FDR) in the interference case now reflects meaningful classification of interference effects, rather than an absence of detection. Overall, the refined analyzer transforms unclassified noise into interpretable interference information, enhancing diagnostic reliability and providing a clearer representation of radar performance under interference conditions.

Velocity Profile Comparison

To further investigate the impact of mutual interference on motion estimation accuracy, velocity profiles were generated for both the baseline and interference scenarios. Each frame's velocity data was extracted from TLV 1 and plotted sequentially to visualize the evolution of detected object velocities over time.

Unlike the other analyses, in this comparison the clutter removal feature was enabled to minimize static background reflections and isolate the motion of dynamic objects more effectively. The remaining configuration parameters, including chirp slope, total frames, and acquisition rate, were kept identical to ensure a fair comparison between baseline and interference conditions.

The following two plots illustrate the difference between baseline and interference conditions. The baseline plot shows a stable and consistent velocity pattern over time, while the interference plot exhibits noticeable fluctuations and irregularities in the detected velocities, confirming the degradation in Doppler measurement accuracy due to interference.

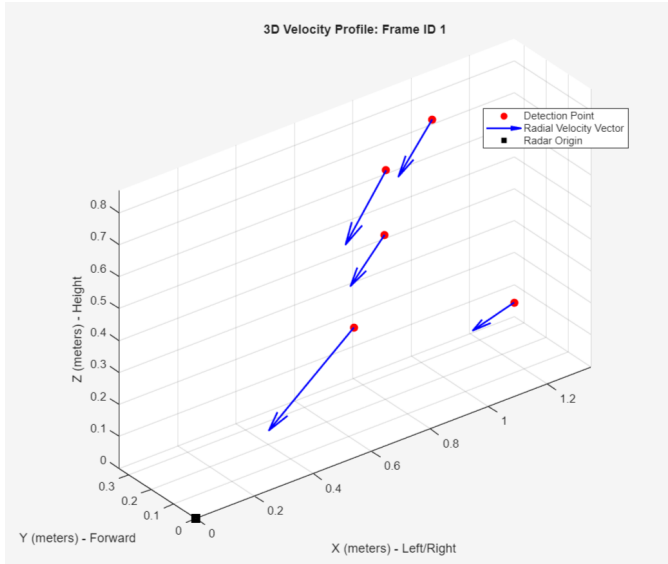


Fig. 6. Velocity profile of baseline condition (clutter removal enabled).

The interference case exhibited non-uniform velocity bands and irregular Doppler shifts, confirming degradation in velocity coherence—a direct indicator of interference-induced distortion in the FMCW Doppler processing chain.

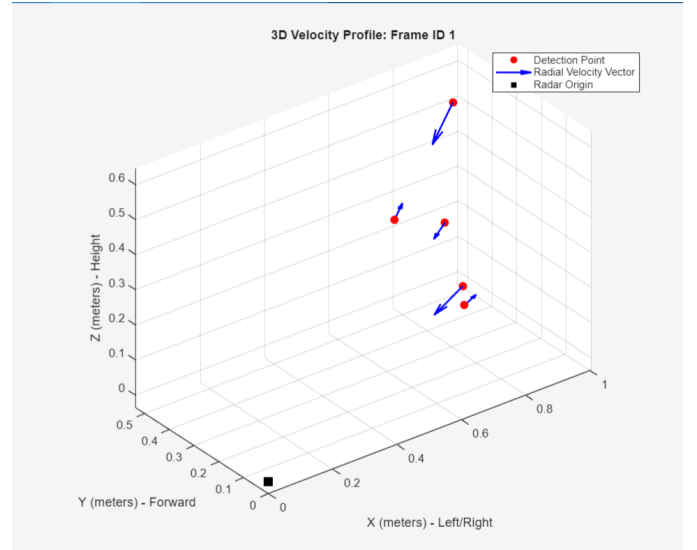


Fig. 7. Velocity profile of interference condition (clutter removal enabled).

Additional Analyses

Beyond this primary setup, a total of nine experiments were conducted by systematically varying chirp slope, total frame count, and clutter removal settings. These additional analyses provided a comprehensive view of how interference characteristics evolve under different radar configurations. Due to space limitations, only the representative configuration (baseline slope = 3, interferer slope = 7, 500 frames) is discussed in detail here. The complete dataset, MATLAB analysis codes, and summarized results (in Excel format) are publicly available at:

The analysis codes and summarized results (in Excel format) are publicly available at:

https://github.com/khair001/Thesis_Radar

Discussion

The experimental results obtained from the observer radar under controlled interference conditions provide strong empirical support for the theoretical framework discussed earlier. Each stage of the analysis—statistical evaluation, clustering refinement, and velocity-domain profiling offers complementary insight into the manifestation of mutual interference in FMCW radar systems.

a) Statistical Observations and Histograms: The increase in the total number of raw detections in the interference-present datasets indicates the injection of spurious reflections and non-coherent energy, consistent with slope-mismatch interference models described in the literature [5]. Broadened SNR and noise histograms further confirm the degradation of signal quality and the rise in background noise floor, a characteristic effect of overlapping chirps from uncoordinated radars [7]. These statistical distortions directly reflect the loss of phase coherence and reduced detection reliability predicted in prior studies.

b) Clustering and Threshold-Based Refinement: The DBSCAN clustering results revealed that under interference conditions, spatial object boundaries became less distinct and cluster densities reduced. Initially, no separate interference cluster was visible; however, applying a dynamic threshold refined the DBSCAN output and exposed well-defined interference clusters. This adaptive thresholding approach demonstrates a practical diagnostic technique for distinguishing between true physical targets and interference-induced detections in processed radar data. Such refinement aligns with the “detect-and-repair” paradigm of interference mitigation discussed in earlier works [7], [9].

c) Velocity Profile and Coherence Analysis: Perhaps the most significant insight emerged from the frame-wise velocity profile analysis. The baseline datasets exhibited stable and coherent Doppler patterns corresponding to consistent target motion, while interference datasets showed scattered and discontinuous velocity traces. This loss of Doppler coherence validates the proposed *Average Directional Variance (ADV)* metric as an effective quantitative indicator of interference-induced corruption. The observed ADV behavior supports theoretical predictions that mutual interference primarily disrupts velocity-domain consistency by introducing non-coherent phase noise [14], [15].

d) Implications and Relevance: Overall, the experiment successfully reproduced key interference artifacts—spurious detections, broadened noise floor, and velocity decoherence—within a real-world dual-radar setup. The results confirm that even without raw ADC access, velocity-domain analysis using processed point cloud data can reveal interference signatures with high reliability. This finding highlights the potential of ADV-based diagnostics as a non-invasive method for real-time interference monitoring in commercial mmWave systems. Furthermore, the methodology provides a foundation for future research on interference-aware signal processing and adaptive radar configuration.

In summary, the results demonstrate strong agreement between theoretical predictions and experimental outcomes. Slope-mismatch interference was effectively characterized through both amplitude (SNR) and phase (velocity) indicators, validating the overall diagnostic framework and reinforcing the critical role of velocity coherence in radar interference analysis.

FUTURE WORK

While this study primarily focused on detecting and characterizing mutual interference through velocity-domain analysis, future work will extend these findings toward interferer localization. The next phase of research will aim to identify the spatial origin of the interference source using the radar’s angle-of-arrival (AoA) capabilities and advanced spatial-temporal processing techniques.

A key objective will be to exploit the IWR6843AOP’s multiple receive antennas to estimate the Angle of Arrival (AoA) of interference components. By analyzing the angular distribution of corrupted detections and correlating them with

the degradation of velocity coherence (as measured by ADV), it will be possible to infer the direction from which interference energy arrives. This approach effectively transforms the interference signature from a nuisance signal into a diagnostic cue for interferer direction finding.

Further development will include the implementation of spectral-spatial correlation, where interference-affected frames are identified via ADV or SNR anomalies and then cross-referenced with their angular intensity profiles. The resulting angular-spectral signatures can be used to build a spatial map of interference activity. Additionally, temporal-angular tracking techniques, such as Kalman filtering or DBSCAN clustering in the angle-time domain, could be employed to monitor moving interferers and estimate their trajectories over time.

Ultimately, the long-term goal of this research is to achieve localization of the interference source through triangulation or multi-radar coordination. By combining AoA estimation from two or more observer radars at known positions, the interferer’s location can be inferred geometrically. Such capability would form the foundation for next-generation radar systems capable not only of detecting interference but also of spatially identifying and mitigating its impact in real time.

V. CONCLUSION

This study examined the effects of mutual interference (MI) in mmWave FMCW automotive radar systems using a dual IWR6843AOP setup. Controlled slope-mismatch experiments revealed that overlapping chirps degrade detection reliability and Doppler coherence. A MATLAB-based analyzer was developed to assess these effects through clustering and threshold-based detection. After applying adaptive threshold calibration, distinct interference clusters were identified that were previously undetectable using static parameters.

Subsequent velocity-domain analysis using the Average Directional Variance (ADV) metric validated the observed loss of velocity coherence and demonstrated the method’s capability for quantifying interference-induced anomalies. The results confirm that the integration of statistical clustering with ADV-based coherence evaluation provides a robust diagnostic framework for characterization of interference.

Overall, this work establishes a solid foundation for future research focused on localizing interference sources and enhancing the reliability of automotive radar perception in complex multi-radar environments.

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